Shahar Berger

Boston, MA | 857-565-6192 | bergershahar10@gmail.com | linkedin.com/in/shahar-berger1 | Portfolio

Medical Device R&D Engineer · Systems · Process · V&V · Authorized to work in the U.S. (no sponsorship required)

SKILLS

Engineering: Systems engineering, V&V, root-cause analysis, design controls, Design for Manufacturing (DFM), test planning, sensor calibration, SolidWorks (CAD), hardware–software integration in regulated environments, Lean Six Sigma Green Belt.

Software & Data: Python (machine learning, image processing, automation), MATLAB (data & signal analysis), RAG / LLM applications (FastAPI, Next.js, Claude & Mistral APIs), data pipelines, React / TypeScript.

Domain: Medical device development (FDA Class II, ISO 14971), optics & medical imaging, regulatory affairs, vendor management & component sourcing, clinical workflow analysis.

PROFESSIONAL EXPERIENCE

R&D Systems Engineer II | Fresenius Medical Care (NxStage), MA — 2025

- Led root-cause investigation of a critical performance failure in a Class II home-hemodialysis device, designing **6 test plans** across 4 subsystems (pump, motor, software, disposable) to isolate a defect from a regulatory-driven material change.
- Ran **100+ tests** and an 18-variable MATLAB analysis to pinpoint the root cause, evaluated solution paths against user experience, regulatory risk, and time-and-development trade-offs, and **recommended a device modification to senior leadership**.
- Maintained formal change-control records and FDA-required design documentation, keeping the program audit-ready throughout.

Product Development Engineer, R&D | GI View Ltd., IL — 2022–2024

- Owned end-to-end development of **3 FDA Class II endoscopy devices** from concept to clearance, driving system design, V&V, and technical trade-offs across R&D, clinical, and regulatory.
- Traced an 80% assembly failure rate to a high-variability manual step; designed a precision fixture with real-time guidance software that lifted production yield **from 20% to 80%** and set quality benchmarks adopted company-wide.
- Led the camera's image-processing and automated sensor-calibration features, lifting **resolution 12.2%** and cutting **color-accuracy error 11× (delta-E 20 → 1.8)** with a custom Python pipeline that replaced subjective manual review with an FDA-submittable tool.
- Built vendor partnerships from scratch: turned a non-responsive CMOS sensor supplier into an active collaborator, and steered a Chinese contract manufacturer through spec definition, quality requirements, and test procedures.
- Directed multiple sensor suppliers and an optical lens-design house, defining part specifications and running market research to source components that met them.
- Ran structured interviews with surgeons and clinical experts, translating clinical needs into engineering requirements; presented progress and trade-offs to C-suite and mentored junior engineers.

Project Engineer | TAU Biomedical Engineering, IL — 2022–2023

- Designed and prototyped a wearable **LiDAR-based motion-capture system** for a 16-participant clinical study, translating clinical requirements into technical specifications.
- Built an **ML classifier** to distinguish stable from unstable movement, plus qualitative and quantitative evaluation methods, turning raw LiDAR signal into the safety and reliability data behind the clinical-trial submission.

Product & Innovation Lead | TAU MedTech Hackathon (Synergy Innovate), IL — 2022–2023

- Founded and ran a healthcare-engineering competition: **18 teams, ~200 participants, 6 clinical challenges**, backed by 5 industry sponsors (Amazon, Google, Microsoft, Philips, IBM); secured hospital data access and coordinated technical mentors from kickoff to finals.

PROJECTS

Prior Authorization Assistant (RAG) | Personal Project — 2026

- Built an end-to-end retrieval-augmented-generation system from scratch (FastAPI, Next.js, Mistral, custom BM25 + vector store, clinical-safety guardrails) to reason through insurance coverage questions; **251 hermetic tests**, 5/5 on an adversarial verdict set.

AI Copilot — Serverless LLM App | Personal Project — 2026

- Designed and shipped an LLM-powered web app in one week (React/TypeScript, Claude API, autonomous agent workflow), spec-first with a public engineering decision record.

EDUCATION & CERTIFICATIONS

M.S., Bioengineering — Northeastern University, MA · GPA 4.0 — 2024–2026

B.S., Biomedical Engineering — Tel Aviv University, IL · Dean's Excellence Award — 2019–2023

Certifications: Embedded Systems & Robotics (Udacity) · Generative AI Fluency (Udacity) · MATLAB Certified (MathWorks) · Optics for Engineers.